

The Quantum Theory and the Electronic Structure of Atoms

Every atom contains electrons. But where exactly are they? How do they behave? For a long time, scientists thought electrons orbited the nucleus like planets around the sun. Then came a series of discoveries that shattered that simple picture. Electrons do not move in neat, predictable paths. They exist in clouds of probability, in discrete energy levels, in shapes described by mathematics rather than simple drawings. Yet most people never learn how this strange quantum world explains the entire periodic table.

Chemistry is not merely a subject taught in classrooms. It is the study of matter at the atomic level. Quantum theory explains why elements behave the way they do — why sodium reacts violently with water, why iron is magnetic, why neon glows red in a sign. Understanding quantum theory means understanding the arrangement of electrons, and that arrangement explains the patterns in the periodic table, the formation of chemical bonds, and the very structure of the world around us.

This reading material explores the contributions of Planck, Bohr, and Schrödinger to quantum theory, the arrangement of electrons in energy levels, subshells, and orbitals, the development of the periodic table by Mendeleev, the relationship between atomic radius and atomic mass, the spdf notation for orbitals, and how to write electronic configurations for common elements.

Part 1: What Is Quantum Theory and Why Does It Matter?

Definition of Quantum Theory

Quantum theory is the branch of physics that explains the behavior of matter and energy at the atomic and subatomic level. It describes electrons not as particles moving in fixed paths but as wave-particle hybrids existing in regions of probability called orbitals.

The Central Idea: Classical Physics Failed at the Atomic Level

Newton's laws work perfectly for baseballs and planets. But they completely fail to explain why atoms emit only specific colors of light, why electrons do not spiral into the nucleus, or why elements have repeating properties. Quantum theory was developed to explain these mysteries.

Why Quantum Theory Matters in Chemistry

Aspect	Explanation
Explains electron arrangement	Quantum theory describes how electrons occupy energy levels, subshells, and orbitals
Predicts chemical bonding	The number and arrangement of valence electrons determine how atoms bond
Explains the periodic table	The patterns in the periodic table directly reflect the quantum structure of atoms
Enables technology	Lasers, transistors, LEDs, MRI machines, and solar cells all rely on quantum principles
Explains atomic spectra	Each element emits unique colors because electrons jump between discrete energy levels

Part 2: The Development of Quantum Theory – Contributions of Planck, Bohr, and Schrödinger

Quantum theory did not emerge overnight. It developed through the work of several scientists who challenged classical physics.

Max Planck – The Birth of Quantum Theory (1900)

At the turn of the 20th century, physicists were puzzled by a simple but frustrating problem: explaining the color of light emitted by hot objects. When you heat a piece of metal, it first glows dull red, then orange, then yellow, then white-hot. Classical physics could predict some of this behavior, but it failed dramatically at high frequencies. The existing theory predicted that hot objects should emit infinite energy in the ultraviolet range – a result that was clearly nonsense, dubbed the "ultraviolet catastrophe." Something was terribly wrong with classical physics. Into this crisis stepped Max Planck, a German physicist who made a radical suggestion that would forever change our understanding of energy.

Aspect	Explanation
Problem	Classical physics could not explain the spectrum of light from hot objects (blackbody radiation). The theory predicted infinite energy at high frequencies, which was impossible.
Planck's proposal	Energy is not continuous but comes in tiny, discrete packets called quanta (singular: quantum). Think of energy like money – you cannot pay with half a cent because there is a smallest unit (one cent). Similarly, energy has a smallest unit (one quantum).
The equation	$E = h \times f$ Where E is the energy of one quantum, h is Planck's constant (6.626×10^{-34} J·s), and f is the frequency of the radiation. This equation means that higher frequency light (like ultraviolet) comes in higher-energy packets.
Significance	This was the first time any physicist suggested that energy could be quantized – that only certain values of energy are allowed. It contradicted centuries of classical physics, which assumed energy was continuous like a smooth flowing river.
Legacy	Planck won the Nobel Prize in Physics in 1918. His constant (h) is now recognized as one of the fundamental constants of the universe, and his quantum idea launched the entire field of quantum mechanics.

Niels Bohr – The Planetary Model with Quantized Orbits (1913)

Less than fifteen years after Planck's discovery, a young Danish physicist named Niels Bohr applied the quantum idea to the atom. At the time, Ernest Rutherford had just discovered that atoms consist of a tiny, dense, positively charged nucleus surrounded by orbiting electrons – like a miniature solar system. But this planetary model had two fatal flaws. First, classical physics predicted that orbiting electrons (which are accelerating) should continuously radiate energy and spiral into the nucleus within a fraction of a second. Since atoms are stable, this clearly did not happen. Second, when atoms were heated, they emitted light only at specific wavelengths (colors), not a continuous rainbow. Bohr realized that Planck's quantum idea could solve both problems if he made a bold assumption: electrons can only exist in certain allowed orbits.

Aspect	Explanation
Problem	Rutherford's model (nucleus with orbiting electrons) could not explain why electrons do not crash into the nucleus (atoms are stable) or why atoms emit only specific colors of light (line spectra) rather than a continuous spectrum.
Bohr's proposal	Electrons exist only in specific, allowed energy levels (orbits) around the nucleus. These are like rungs on a ladder – you can stand on a rung, but you cannot stand between rungs.
Key ideas	1. While in an allowed orbit, an electron does NOT radiate energy (contradicting classical physics). 2. Electrons can jump between levels by absorbing or emitting a photon (light) of exact energy. The energy of the photon equals the difference between the two energy levels ($\Delta E = E_{\text{final}} - E_{\text{initial}}$). 3. The lowest energy level ($n=1$) is called the ground state; higher levels ($n=2, 3, 4, \dots$) are excited states.
Limitations	Bohr's model worked perfectly for hydrogen (which has only one electron) and for ions with one electron (He^+ , Li^{2+}). However, it completely failed for atoms with more than one electron. It could not predict their spectra or

Aspect	Explanation
	explain their chemical properties.
Significance	Despite its limitations, Bohr's model introduced the revolutionary concept of discrete energy levels in atoms. It showed that quantum rules apply to electrons in atoms and explained why atoms are stable. Bohr won the Nobel Prize in Physics in 1922.

Erwin Schrödinger – The Quantum Mechanical Model (1926)

Bohr's model, while successful for hydrogen, was clearly incomplete. It treated electrons as tiny planets moving in well-defined circular orbits – a picture that turned out to be too simple. Experiments in the 1920s showed that electrons behave like waves; they can produce interference patterns (diffraction) just like light. A new kind of theory was needed, one that could account for the wave-like nature of electrons and work for all atoms, not just hydrogen. Enter Erwin Schrödinger, an Austrian physicist who, in a burst of creativity in 1926, developed a wave equation that completely transformed our understanding of the atom. Instead of asking "Where is the electron exactly?" Schrödinger asked "What is the probability of finding the electron in a given region of space?"

Aspect	Explanation
Problem	Bohr's model could not explain atoms with multiple electrons, and it completely ignored the newly discovered wave-like behavior of electrons. Experiments showed that electrons produce diffraction patterns – something only waves should do.
Schrödinger's proposal	Electrons are not particles moving in fixed paths like planets. Instead, they are wave-functions (represented by the Greek letter ψ , psi) described by the Schrödinger equation. The electron does not have a single, definite position; its location is spread out like a cloud of probability.
Key idea	Instead of exact positions, electrons exist in orbitals – three-dimensional

Aspect	Explanation
	regions of space where there is a high probability (approximately 90% or greater) of finding the electron. You can never know exactly where an electron is, only where it is likely to be.
The equation	$\hat{H}\psi = E\psi$ (the time-independent Schrödinger equation) Where $\hat{H}$ is the Hamiltonian operator (representing total energy), ψ is the wave function, and E is the energy of the system. Solving this equation for an atom gives the allowed energy levels and the shapes of orbitals.
Significance	Schrödinger's model forms the foundation of modern quantum mechanics. Unlike Bohr's model, it works for all atoms (with reasonable approximations). It gives us the four quantum numbers (n , l , m_l , m_s) that completely describe an electron's state. It explains the shapes of s, p, d, and f orbitals, and it provides the theoretical basis for the entire periodic table.
Legacy	Schrödinger won the Nobel Prize in Physics in 1933, sharing it with Paul Dirac. His wave equation is one of the most important equations in all of physics, and his probability interpretation of the electron (developed with Max Born) remains the standard view today.

Part 3: The Quantum Mechanical Model – Energy Levels, Subshells, Orbitals, and Quantum Numbers

In the quantum mechanical model developed by Schrödinger and others, electrons do not move in fixed paths like planets. Instead, they are described by **wave functions** that give the probability of finding an electron in a particular region of space. These regions are called **orbitals**. To completely describe the state and behavior of every electron in an atom, scientists use **four quantum numbers**. These quantum numbers arise directly from solving the Schrödinger equation and determine the arrangement of electrons into energy levels, subshells, and orbitals.

The Principal Quantum Number (n) – Main Energy Levels

The first and most important quantum number is the **principal quantum number**, denoted by **n**. It describes the main energy level (or shell) occupied by an electron. The value of n tells you two things: the energy of the electron and its most probable distance from the nucleus. An electron with $n = 1$ is closer to the nucleus and has lower energy than an electron with $n = 2$. A higher value of n means the electron is, on average, farther from the nucleus and has higher energy.

The principal quantum number can assume any positive integer value: $n = 1, 2, 3, 4$, and so on. Each value of n corresponds to a main energy level. The maximum number of electrons that can fit into a given energy level is given by the formula **$2n^2$** .

n Value	Name	Maximum Electrons ($2n^2$)
$n = 1$	First energy level	$2 \times 1^2 = 2$ electrons
$n = 2$	Second energy level	$2 \times 2^2 = 8$ electrons
$n = 3$	Third energy level	$2 \times 3^2 = 18$ electrons
$n = 4$	Fourth energy level	$2 \times 4^2 = 32$ electrons

Every subshell in an atom is referred to by a code that combines n and a letter, such as 1s, 2s, 2p, 3s, 3p, 3d, and so on. The integer in this code is the principal quantum number n .

The Angular Momentum (Azimuthal) Quantum Number (ℓ) – Subshells and Shapes

The second quantum number is the **angular momentum quantum number**, denoted by ℓ (lowercase L). While n tells you the size and energy of an electron's orbital, ℓ tells you its **shape**. The letter in the subshell code (s, p, d, or f) directly corresponds to the value of ℓ .

The angular momentum quantum number can only assume integer values from 0 up to $(n - 1)$. For a given value of n , ℓ can be 0, 1, 2, ..., up to $(n - 1)$. Each value of ℓ corresponds to a specific subshell with a characteristic shape.

ℓ Value	Subshell Letter	Shape Description	First Appears at $n =$
$\ell = 0$	s	Sphere	$n = 1$
$\ell = 1$	p	Dumbbell (two lobes)	$n = 2$
$\ell = 2$	d	Cloverleaf (four lobes)	$n = 3$
$\ell = 3$	f	Complex (multilobed)	$n = 4$

Thus, the subshell 1s has $n = 1$ and $\ell = 0$. The subshell 2p has $n = 2$ and $\ell = 1$. The subshell 3d has $n = 3$ and $\ell = 2$. The subshell 4f has $n = 4$ and $\ell = 3$.

The Magnetic Quantum Number (m_ℓ) – Orbital Orientation

The third quantum number is the **magnetic quantum number**, denoted by m_ℓ . While ℓ describes the shape of an orbital, m_ℓ specifies its **orientation in space**. For a given p subshell ($\ell = 1$), for example, there are three different orientations: along the x-axis, the y-axis, and the z-axis. These correspond to the p_x , p_y , and p_z orbitals.

The magnetic quantum number can only assume integer values from $-\ell$ to $+\ell$, including zero. The number of possible m_ℓ values for a given ℓ tells you how many orbitals exist in that subshell. Each orbital can hold a maximum of two electrons.

ℓ Value	Subshell	Possible m_ℓ Values	Number of Orbitals	Orbital Labels (Examples)
$\ell = 0$	s	0	1	s orbital
$\ell = 1$	p	-1, 0, +1	3	p_x , p_y , p_z
$\ell = 2$	d	-2, -1, 0, +1, +2	5	d_{xy} , d_{xz} , d_{yz} , $d_{x^2-y^2}$, d_{z^2}
$\ell = 3$	f	-3, -2, -1, 0, +1, +2, +3	7	Various complex orientations

The Spin Quantum Number (m_s) – Electron Spin

The fourth quantum number is the **spin quantum number**, denoted by m_s . The first three quantum numbers (n , ℓ , m_ℓ) describe the orbital that an electron occupies. However, two electrons can occupy the same orbital. How are they distinguished? They must have opposite **spins**.

The spin quantum number specifies the direction in which an electron is spinning on its axis. It can only have one of two possible values: $+\frac{1}{2}$ (called spin-up, represented by $\uparrow$) or $-\frac{1}{2}$ (spin-down, represented by $\downarrow$).

This leads to the **Pauli exclusion principle**, which states that no two electrons in the same atom can have the same set of all four quantum numbers (n , ℓ , m_ℓ , m_s). In practical terms, this means that:

1) each orbital can hold a maximum of **two electrons**

and 2) those two electrons must have **opposite spins** (one $+\frac{1}{2}$ and one $-\frac{1}{2}$).

Summary Table: Energy Levels, Subshells, and Orbitals

Main Energy Level (n)	Available Subshells	Orbitals Present	Total Orbitals	Maximum Electrons
n = 1	s	1s	1	2
n = 2	s, p	2s, 2px, 2py, 2pz	4	8
n = 3	s, p, d	3s, 3p (three), 3d (five)	9	18
n = 4	s, p, d, f	4s, 4p (three), 4d (five), 4f (seven)	16	32

Part 4: Describing a Model of Discrete Energy Levels – spdf Notation

The spdf notation is a shorthand way to write the electronic configuration of an atom. It shows which subshells contain electrons and how many electrons are in each.

Format of spdf Notation

n subshell superscript

Where:

- **n** = main energy level (number)
- **subshell** = letter (s, p, d, or f)
- **superscript** = number of electrons in that subshell

Example: Hydrogen (H)

Information	Value
Atomic number	1 (one electron)
Electronic configuration (spdf)	$1s^1$
Meaning	Energy level 1, s subshell, 1 electron

Example: Helium (He)

Information	Value
Atomic number	2 (two electrons)
Electronic configuration (spdf)	$1s^2$
Meaning	Energy level 1, s subshell, 2 electrons (full)

Example: Carbon (C)

Information	Value
Atomic number	6 (six electrons)
Electronic configuration (spdf)	$1s^2 2s^2 2p^2$
Meaning	$1s^2$ (first level full), $2s^2$ (second level s full), $2p^2$ (two electrons in p subshell)

Part 5: Writing Electronic Configurations Using Quantum Theory Principles

To write electronic configurations correctly, you must follow three key principles.

Principle 1: Aufbau Principle (Building Up)

Electrons fill the lowest energy orbitals first before occupying higher energy orbitals.

Order of Filling

$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \rightarrow 4d \rightarrow 5p \rightarrow 6s$
 $\rightarrow 4f \rightarrow 5d \rightarrow 6p \rightarrow 7s \rightarrow 5f \rightarrow 6d \rightarrow 7p$

Explanation

Follow the diagonal rule (see diagram below)

Principle 2: Pauli Exclusion Principle

No two electrons in the same atom can have the same set of four quantum numbers. In practical terms: **each orbital can hold a maximum of two electrons, and they must have opposite spins.**

Rule

Meaning

Maximum 2 electrons per orbital

One spin up ($\uparrow$), one spin down ($\downarrow$)

Opposite spins

Electrons in the same orbital must be paired with opposite magnetic orientation

Principle 3: Hund's Rule

When electrons occupy orbitals of the same energy (e.g., three p orbitals), they first fill each orbital singly with parallel spins before pairing up.

Correct Filling

$\uparrow \uparrow \uparrow$ (all separate)

Then $\uparrow \downarrow \uparrow \uparrow$ (pair after all are half-filled)

Incorrect Filling

$\uparrow \downarrow \uparrow$ (paired early)

Worked Example: Writing Electronic Configurations

Element	Atomic Number	Electronic Configuration (spdf)	Shorthand (Noble Gas Core)
Hydrogen (H)	1	$1s^1$	[He] (no, H is simpler)
Helium (He)	2	$1s^2$	$1s^2$
Lithium (Li)	3	$1s^2 2s^1$	[He] $2s^1$
Beryllium (Be)	4	$1s^2 2s^2$	[He] $2s^2$
Boron (B)	5	$1s^2 2s^2 2p^1$	[He] $2s^2 2p^1$
Carbon (C)	6	$1s^2 2s^2 2p^2$	[He] $2s^2 2p^2$
Nitrogen (N)	7	$1s^2 2s^2 2p^3$	[He] $2s^2 2p^3$
Oxygen (O)	8	$1s^2 2s^2 2p^4$	[He] $2s^2 2p^4$
Fluorine (F)	9	$1s^2 2s^2 2p^5$	[He] $2s^2 2p^5$
Neon (Ne)	10	$1s^2 2s^2 2p^6$	[Ne] (full octet)
Sodium (Na)	11	$1s^2 2s^2 2p^6 3s^1$	[Ne] $3s^1$
Magnesium (Mg)	12	$1s^2 2s^2 2p^6 3s^2$	[Ne] $3s^2$

Element	Atomic Number	Electronic Configuration (spdf)	Shorthand (Noble Gas Core)
Aluminum (Al)	13	$1s^2 2s^2 2p^6 3s^2 3p^1$	$[\text{Ne}] 3s^2 3p^1$
Chlorine (Cl)	17	$1s^2 2s^2 2p^6 3s^2 3p^5$	$[\text{Ne}] 3s^2 3p^5$
Argon (Ar)	18	$1s^2 2s^2 2p^6 3s^2 3p^6$	$[\text{Ar}]$

Exceptions to the Rules (Memorize These)

Element	Expected Configuration	Actual Configuration	Reason
Chromium (Cr)	$[\text{Ar}] 4s^2 3d^4$	$[\text{Ar}] 4s^1 3d^5$	Half-filled d subshell (d^5) is extra stable
Copper (Cu)	$[\text{Ar}] 4s^2 3d^9$	$[\text{Ar}] 4s^1 3d^{10}$	Completely filled d subshell (d^{10}) is extra stable

Part 6: How Mendeleev Developed the Periodic Table

Before Mendeleev, chemists had no organized way to classify elements. Several scientists attempted to find patterns, but Mendeleev was the first to produce a truly useful periodic table.

Dmitri Mendeleev (1834-1907)

Aspect	Explanation
Background	Russian chemist and professor

Aspect	Explanation
Problem in his time	63 elements were known, but there was no system to organize them; properties seemed random
Mendeleev's method	He wrote each element's name, atomic mass, and properties on a card. He arranged the cards in order of increasing atomic mass and looked for repeating (periodic) patterns

Mendeleev's Key Innovations

Innovation	Explanation	Example
Arranged by atomic mass	Elements were ordered by increasing atomic mass	Li (7), Be (9), B (11), C (12), etc.
Identified periodic law	Properties of elements repeat at regular intervals	Every 8th element had similar properties (octaves)
Left gaps for undiscovered elements	He predicted elements that had not yet been discovered	Predicted eka-aluminum (gallium), eka-silicon (germanium)
Corrected wrong atomic masses	He reversed some elements if properties did not fit	Tellurium (Te) placed before iodine (I) even though Te has higher atomic mass
Predicted properties	He predicted the properties of missing elements with remarkable accuracy	Predicted density, melting point, and formula of eka-silicon (germanium)

Mendeleev vs. Modern Periodic Table

Feature	Mendeleev's Table (1869)	Modern Periodic Table
Ordering principle	Increasing atomic mass	Increasing atomic number (protons)
Number of elements	63 known + gaps for predicted	118 confirmed
Noble gases	Not present (not discovered yet)	Group 18
Actinides and lanthanides	Not present	Placed separately below
Explanation of periodicity	Not known (quantum theory did not exist yet)	Explained by electron configuration (valence electrons)

Why Mendeleev's Table Was Revolutionary

Reason	Explanation
Predicted discovery	His gaps told scientists exactly what to look for
Predicted properties	When gallium and germanium were discovered, their properties matched Mendeleev's predictions almost perfectly
Organized knowledge	The table turned a random list of elements into a powerful predictive tool
Inspired quantum theory	The search for the underlying cause of periodicity led to the discovery of atomic number and electron configuration

Part 7: Diagram of the Periodic Table – Atomic Radius and Atomic Mass

The periodic table reveals clear trends in atomic radius (size of atoms) and atomic mass. Understanding these trends helps explain the physical and chemical behavior of elements.

Relationship Between Atomic Radius and Atomic Mass

- **Across a period (left to right):** Atomic radius decreases; atomic mass increases
- **Down a group (top to bottom):** Atomic radius increases; atomic mass increases

Diagram of Atomic Radius Trends (Draw these)

Period 2 (Li to Ne): Atomic Radius (pm)

Li	Be	B	C	N	O	F	Ne
152	111	88	77	70	66	64	67

(Largest at left; decreases across the period; Ne is slightly larger due to electron repulsion)

Group 1 (Alkali Metals): Atomic Radius (pm)

Li	Na	K	Rb	Cs
152	186	227	248	265

(Increases down the group as each step adds a new energy level)

Complete Table: Trends in Periods and Groups

Trend	Across a Period (Left → Right)	Down a Group (Top → Bottom)
Atomic radius	Decreases (increased nuclear charge pulls electrons inward)	Increases (new energy levels added)
Atomic mass	Increases (more protons and neutrons)	Increases (more protons and neutrons)
Explanation	Same energy level (n) but more protons → stronger pull → smaller atom	New energy level (higher n) → electron farther from nucleus → larger atom

Why Atomic Radius Decreases Across a Period

Step	Explanation
1	As you move left to right, the number of protons increases (higher atomic number)
2	More protons means a stronger positive charge in the nucleus
3	The electrons are added to the same main energy level (n)
4	The stronger nuclear pull draws the electron cloud inward
5	Result: The atom becomes smaller (radius decreases)

Example: Period 2 Atomic Radii

Element	Li	Be	B	C	N	O	F	Ne
Atomic radius (pm)	152	111	88	77	70	66	64	67
Decreasing Trend	→	→	→	→	→	→	→	(Ne is slightly larger due to electron repulsion)

Example: Group 1 Atomic Radii

Element	Li	Na	K	Rb	Cs
Atomic radius (pm)	152	186	227	248	265
Trend	↓	increases	↓	↓	(each step adds a new energy level)

Part 8: Explaining Periods and Groups Using Electronic Configuration

The periodic table is organized exactly according to the electronic configurations of atoms. The position of an element tells you its electron arrangement.

Periods (Horizontal Rows)

The **period number** tells you the highest main energy level (n) that contains electrons.

Period	Highest n	Electronic Configurations End With	Number of Elements
Period 1	n = 1	1s ¹ , 1s ²	2
Period 2	n = 2	2s ¹ , 2s ² , 2p ¹ ... 2p ⁶	8
Period 3	n = 3	3s ¹ , 3s ² , 3p ¹ ... 3p ⁶	8
Period 4	n = 4	4s ¹ ... 4p ⁶ (and 3d)	18
Period 5	n = 5	5s ¹ ... 5p ⁶ (and 4d)	18

Period	Highest n	Electronic Configurations End With	Number of Elements
Period 6	n = 6	6s ¹ ... 6p ⁶ (and 4f, 5d)	32
Period 7	n = 7	7s ¹ ... 7p ⁶ (and 5f, 6d)	32 (incomplete)

Groups (Vertical Columns)

Elements in the same group have the **same number of valence electrons** (electrons in the outermost main energy level). This explains why they have similar chemical properties.

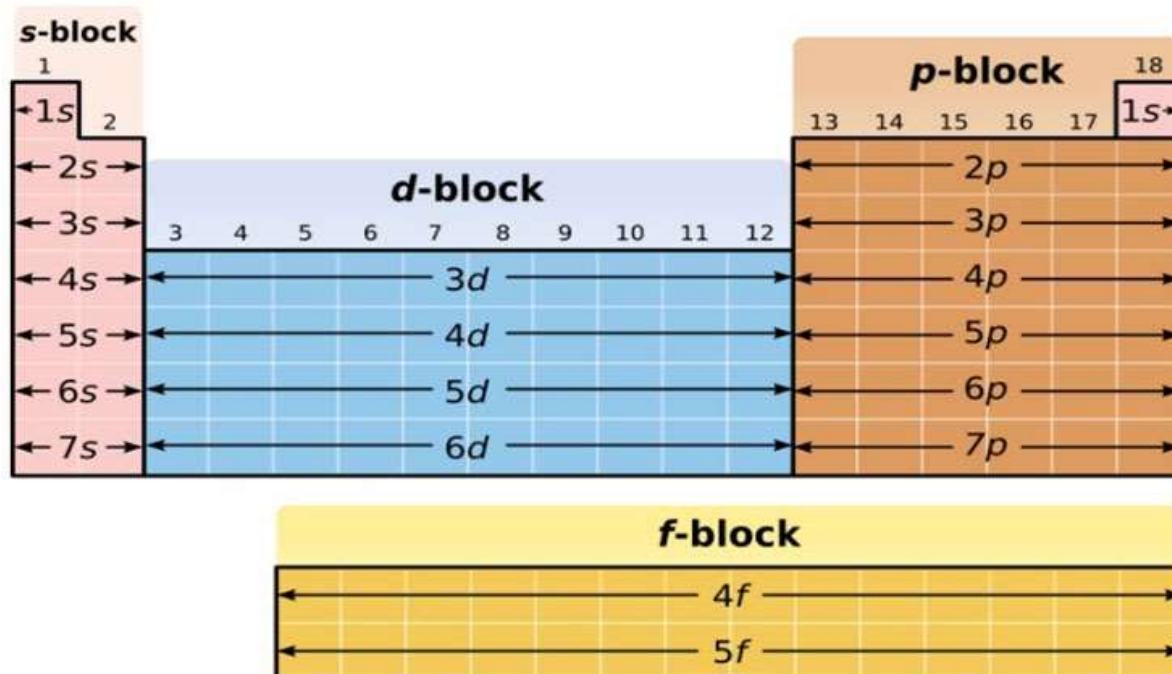
Group Number	Valence Electrons	Electronic Configuration of Valence Shell
Group 1 (Alkali metals)	1 valence electron	ns ¹
Group 2 (Alkaline earth metals)	2 valence electrons	ns ²
Group 13 (Boron group)	3 valence electrons	ns ² np ¹
Group 14 (Carbon group)	4 valence electrons	ns ² np ²
Group 15 (Nitrogen group)	5 valence electrons	ns ² np ³
Group 16 (Chalcogens)	6 valence electrons	ns ² np ⁴
Group 17 (Halogens)	7 valence electrons	ns ² np ⁵
Group 18 (Noble gases)	8 valence electrons (except He with 2)	ns ² np ⁶

Block Structure of the Periodic Table

The periodic table is divided into four blocks (s, p, d, f) based on which subshell is being filled with electrons.

Block	Subshell Being Filled	Location on Periodic Table	Example
s-block	s orbital	Groups 1-2 (and He)	Na: [Ne] $3s^1$
p-block	p orbitals	Groups 13-18	Cl: [Ne] $3s^2 3p^5$
d-block	d orbitals	Transition metals (Groups 3-12)	Fe: [Ar] $4s^2 3d^6$
f-block	f orbitals	Lanthanides and actinides (below main table)	U: [Rn] $7s^2 5f^3$

The Periodic Table Blocks



Predicting Electronic Configuration from the Periodic Table

You can determine an element's electronic configuration simply by locating it on the periodic table.

Step	Explanation	Example (Chlorine, Cl)
1	Find the period (row)	Period 3 → highest $n = 3$
2	Find the block	p-block (Group 17)
3	Find the position within the block	Third element in p-block of Period 3
4	Write core configuration up to previous noble gas	[Ne] (neon, Period 2)
5	Add electrons for period and block	$3s^2 3p^5$
6	Complete configuration	[Ne] $3s^2 3p^5$

Key Terms

Term	Definition
Quantum theory	The theory that explains the behavior of matter and energy at the atomic and subatomic level
Quantum	A discrete, indivisible packet of energy
Energy level (shell)	A region around the nucleus where electrons have a specific energy, designated by n
Subshell	A subdivision of an energy level (s , p , d , f)
Orbital	A region of space with a high probability of finding an electron; holds up to 2 electrons
Aufbau principle	Electrons fill the lowest energy orbitals first
Pauli exclusion principle	No two electrons in an atom can have the same set of four quantum numbers; each orbital holds max 2 electrons with opposite spins
Hund's rule	Electrons fill orbitals singly with parallel spins before pairing
Valence electrons	Electrons in the outermost main energy level of an atom
Atomic radius	Half the distance between the nuclei of two identical atoms bonded together
Periodic law	The properties of elements are periodic functions of their atomic number

Review Questions

1. Describe the contributions of Max Planck, Niels Bohr, and Erwin Schrödinger to the development of quantum theory. For each scientist, state their key idea and why it was significant.
2. Draw a labeled diagram showing the discrete energy levels of an atom using spdf notation for the orbitals. Your diagram must include the first three main energy levels ($n = 1, 2, 3$) and show the s, p, and d subshells with their correct number of orbitals.
3. Write the full electronic configuration (spdf notation) for each of the following elements:
 - (a) Nitrogen (N, atomic number 7)
 - (b) Sodium (Na, atomic number 11)
 - (c) Chlorine (Cl, atomic number 17)
 - (d) Calcium (Ca, atomic number 20)
4. Explain how Mendeleev developed the periodic table of elements. Your answer must include his ordering principle (atomic mass), how he identified periodic patterns, what he did with gaps, and at least one example of a property he predicted correctly.
5. Draw a diagram of the periodic table showing the trend in atomic radius across a period (left to right) and down a group (top to bottom). Your answer must include:
 - A labeled diagram (with arrows showing increasing or decreasing radius)
 - An explanation for why atomic radius decreases across a period
 - An explanation for why atomic radius increases down a group
 - Specific examples from Period 2 and Group 1
6. Explain the relationship between an element's position in the periodic table (period and group) and its electronic configuration. Your answer must address:
 - What the period number tells you about main energy levels
 - What the group number tells you about valence electrons
 - How the block (s, p, d, f) tells you which subshell is being filled

- 7.** Apply the three principles of quantum theory (Aufbau principle, Pauli exclusion principle, and Hund's rule) to determine the electronic configuration of oxygen (O, atomic number 8). Show your step-by-step reasoning and write the final configuration in spdf notation.
- 8.** Predict the electronic configuration of an element located in Period 4, Group 2, s-block. Write the full spdf notation and the shorthand (noble gas core) notation.
- 9.** Explain why elements in the same group have similar chemical properties. Use the concept of valence electrons and electronic configuration in your answer. Provide one example of two elements in the same group.
- 10.** Thinking question: Mendeleev placed tellurium (atomic mass 127.6) before iodine (atomic mass 126.9) even though tellurium has a higher atomic mass. Explain why he did this. How does the modern periodic table resolve this issue?